



Team Omega

Audit of

Backed AtomicSwapForwarder Contract

Final Report

July 14, 2026

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Summary

This document is the result of an audit by Team Omega for Backed Finance of an update of their Atomic Swap contract suite with a “forwarder” contract that allows a set of whitelisted contracts to execute trades on behalf of users.

We found **no high severity issue** - these are issues that can lead to a loss of funds, and are essential to fix. We classified **no** issues as “medium” - these are issues we believe you should definitely address. We did classify **2** issues as “low”, and an additional 2 issues were classified as “info” - we believe the code would improve if these issues were addressed.

On June 23, the Backed Team submitted a new commit, which added an expiry date to the authorization signature. We reviewed these changes.

Of the issues we noted, none were fully addressed.

Severity	Number of issues	Number of resolved issues
High	0	0
Medium	0	0
Low	2	0
Info	2	0

Scope of the Audit

The audit concerns an upgrade of the Backed Swap Contracts, specifically the following PR:

<https://github.com/backed-fi/swap-contract/pull/6>

Specifically, this PR seeks to merge commit

9d7840dc4ab0da7d16cf54922af5393aa54ab6e6

into the current main branch at commit, which we reviewed previously:

2eb1b6c1b1b2e6a123d5defb997533b90a54e4bd

The audit focuses on the changes to the following files:

contracts/swap/AtomicSwapForwarderUpgradeable.sol
contracts/swap/IAtomicSwapForwarder.sol

In November 2025, we audited a previous version of this code. That audit report is available

<https://github.com/OmegaAudits/audits/blob/main/202511-Backed-atomic-swap.pdf>

Resolution

The issues were subsequently addressed on June 23 in the following commit:

e5cf0c111550ee971d596d5414cd9b7217b2a953

We reviewed these new changes and noted our observation below, for each issue.

The code was then merged into the main branch in commit

058913ea8005af27dd1d6c06a6f44dcef2f7d9f5

Methods Used

The contracts were compiled, deployed, and tested in a test environment. We ran static analysis tools and AI tools.

Code Review

We manually inspected the source code to identify potential security flaws. We also inspected the context in which the contracts were developed.

Automatic analysis

We have used static analysis tools to detect common potential vulnerabilities. No high severity issues were identified with the automated processes. Some low severity issues were found, and we have included them below in the appropriate parts of the report.

Disclaimer

The audit makes no statements or warranties about utility of the code, safety of the code, suitability of the business model, regulatory regime for the business model, or any other statements about fitness of the contracts to purpose, or their bug free status. The audit documentation is for discussion purposes only.

Severity definitions

High	Vulnerabilities that can lead to loss of assets or data manipulations.
Medium	Vulnerabilities that are essential to fix, but that do not lead to assets loss or data manipulations
Low	Issues that do not represent direct exploit, such as poor implementations, deviations from best practice, high gas costs, etc
Info	Matters of opinion

Findings

G1. Test Coverage is not complete [low] [not resolved]

Although most critical paths are covered, test coverage is not complete:

all files swap/

96.92% Statements 63/65 86.76% Branches 59/68 90.91% Functions 20/22 97.7% Lines 85/87

File		Statements	Branches	Functions	Lines
AtomicSwapForwarderUpgradeable.sol		93.75%	30/32	80%	24/30
AtomicSwapUpgradeable.sol		100%	33/33	92.11%	35/38
IAAtomicSwap.sol		100%	0/0	100%	0/0
IAAtomicSwapForwarder.sol		100%	0/0	100%	0/0

What is missing is coverage of pausability, and the re-entrancy decorator.

Recommendation: Have complete test coverage, to confirm behavior but also to make sure that in future iterations, the behavior stays the same

Severity: Low - all complex paths have tests

Resolution: The issue was not resolved, and coverage percentages slightly diminished after the update.

G2. Documentation is not consistent with the code [low] [not resolved]

The documentation in `FORWARDER.md` lists a number of use cases, among which are "custodial counterparty trading on behalf of many end-clients, or a routing service that executes RFQ-style quotes for users " and "gasless transactions".

This strongly suggests that there is a decoupling between the funder of the swap and the signer/submitter of the swap tx. But this is not strictly true: the `msg.sender` of the transaction that is required to be the same as the wallet that holds the tokens to fund the tx.

Recommendation: Fix the documentation to reflect the actual behavior of the forwarder contract

Severity: Low

Resolution: The issue was not resolved

G3. `IAAtomicSwapForwarderUpgradeableInterface` is incomplete [info] [partially addressed]

The interface is meant to represent the external surface of `AtomicSwapForwarderUpgradeable`, but it omits view helpers that are implemented on the contract and referenced elsewhere (e.g., docs/tests), specifically:

- `hashProxyAuthorization(bytes32, address)`
- the `atomicSwap()` getter

Recommendation: Consider adding these functions to the interface so integrators can depend on `IAAtomicSwapForwarder` without needing the concrete ABI.

Severity: Info

Resolution: The interface was extended, but does not contain the getter functions for the public variables in `AtomicSwapForwarderUpgradeable`

G4. There is no smooth signer migration path [info] [not resolved]

Currently, the signer can be changed using the `setSigner` function. Calling this function will immediately change the signer, and invalidate all quotes previously signed by the signer.

This behavior is fine if the signer is compromised, but may be inconvenient during normal operations in which you may want to change the signer. For example, given that the signing is a live and probably partly automated process, the account may be protected by a single private key, and in that case, key rotation may be, or may become, part of standard operating procedures.

Recommendation: Consider implementing a mechanism that allows for setting a new signer without invalidating previous quotes.

Severity: Info

Resolution: The issue was not addressed